

Compare Linear and Polynomial Regression using Python

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
from sklearn.metrics import mean_squared_error, r2_score
X = np.array([1, 2, 3, 4, 5, 6, 7, 8]).reshape(-1, 1)
y = np.array([2, 5, 10, 17, 26, 37, 50, 65])
linear_model = LinearRegression()
linear_model.fit(X, y)
y_linear = linear_model.predict(X)
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X)

poly_model = LinearRegression()
poly_model.fit(X_poly, y)

y_poly = poly_model.predict(X_poly)
linear_mse = mean_squared_error(y, y_linear)
linear_r2 = r2_score(y, y_linear)
poly_mse = mean_squared_error(y, y_poly)
poly_r2 = r2_score(y, y_poly)
print("Comparison of Linear and Polynomial Regression")
print("-----")

print("\nLinear Regression:")
```

```
print("MSE:", linear_mse)
print("R2 Score:", linear_r2)

print("\nPolynomial Regression:")
print("MSE:", poly_mse)
print("R2 Score:", poly_r2)

plt.scatter(X, y, label="Actual Data")
plt.plot(X, y_linear, label="Linear Regression")
plt.plot(X, y_poly, label="Polynomial Regression")

plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear vs Polynomial Regression")
plt.legend()
plt.show()
```

OUTPUT:

... Linear Regression:
MSE: 21.0
R2 Score: 0.9529411764705882

Polynomial Regression:
MSE: 4.50636792107503e-29
R2 Score: 1.0

